

Thesis Report Guide

Title Page: The format of the title page will be given as part of a template.

Approval Page: It is a copy of title page but it contains the name and the signature of the thesis advisor.

Abstract: The abstract part is a brief summary of your thesis that concisely describes the content and scope of the project and it identifies the project's objective, methodology and the conclusions. It will be at least 150 words and it will not be more than 300 words. It will be in separate page in the thesis document.

Acknowledgements: The acknowledgement is optional.

Table of Contents: It lists all material within the thesis from the beginning. It must conform to the example provided at the end of this guide.

List of Figures: It includes all figures within the thesis based on the example given in this document.

List of Tables: It includes all tables within the thesis by providing the table number and the title of the table.

Body of Thesis: The compulsory layout of the body of the thesis is given at the next page. *Note that each chapter in the body section will start with a new page and there will be 6 chapters in your thesis.*

References: References must conform to the examples provided at the end of this document.

Appendices: It is an optional section that lists all necessary appendices.

Body of the Thesis

1. Introduction

1.1. Problem Description and Motivation (*revised from PSD/ADD documents*)

Provide general description of the problem and motivation of the study in multiple paragraphs. Background and/or context for understanding the nature of the problem should be provided. You should provide answers to the following questions:

- What is the motivation for this project?
- Why have you done this project?
- Is the project important or worthwhile?
- What have you done? (very briefly)

1.2. Main Goal and Objectives of the Project (*revised from PSD document*)

Provide the main goal of the project, and the objectives to achieve that goal. A goal is an achievable outcome that is generally broad and longer term while an objective is shorter term and defines clear and measurable actions to achieve the overall goal. After defining the main goal, provide a bulleted list of all objectives of the project. All objectives should be clear and specific, measurable, and achievable.

2. Definition of the Project

2.1. Scope of the Project (*revised from PSD/ADD*)

Define your scope precisely and completely. For example, if you have implemented a particular part of a system, explain which parts are in the scope of your project and which parts are out of its scope.

If your project is based on another project (e.g., a previous student project, an open source project, a completed or ongoing project of your supervisor etc.), clearly describe the relationship between them; and specify all required inputs and outputs from the reference project or work.

You should list all constraints and/or limits of the project clearly. You should also discuss any assumptions related to your project. For example, you may assume that there are no more than a thousand simultaneous users for your online multi-user software. You should try to clear up as many assumptions as possible.

2.2. Success Factors (*revised from PSD*)

Describe how success of your project is measured. Specifically, for each objective given in Section 1.2, describe the key performance indicators that show that you have satisfied the requirements.

2.3. Professional Considerations (*revised from PSD*)

This section should include proper explanations for all items listed below:

Methodological considerations/engineering standards: Include all methodological standards and/or language/notational standards that will be used (such as GANTT charts, UML diagrams, Source Code Control via Git/Subversion/etc, IEEE standards, ...). Explain each related item with proper illustrations, i.e., figures, tables.

Realistic Constraints: In addition to the traditional technological and economic considerations fundamental to the design of software and hardware components and systems, a modern engineer has become increasingly concerned with the broader considerations of realistic constraints which are particularly related to the better-off today's society and quality of life. In your project design, you have to be imaginative and ingenious enough to anticipate potentially hazardous situations and all the factors relating to the project outcome and make the best design decision to address those realistic constraint issues. Specifically, in your document, you should consider the following 6 aspects: i) economical, ii) environmental, iii) ethical, iv) health and safety, v) sustainability and vi) social. If your project does not have any consideration in one of these aspects, clearly describe the reason.

Legal considerations, e.g. required permissions if the developed product should come to market, including licenses, medical, financial and ethical permissions.

2.4.Literature Survey / Related Work (*revised from ADD*)

Related studies in the literature are presented in this chapter. The differences of the project are also presented in this chapter.

3. System Design and Software Architecture

Note that this section will be different based on the type of the project.

(For Software Oriented Projects)

3.1. Project Requirements (*revised ADD*)

3.1.1. Functional Requirements

3.1.2. Nonfunctional Requirements

3.2. System Design (*revised from ADD*)

3.2.1. UML Use case Diagram(s) for the main use cases

3.2.2. UML Class and/or Database ER diagram(s)

3.2.3. User Interface (Preliminary version)

3.2.4. Test Plan

3.3.Software Architecture (*revised from ADD*)

Main aspects of data flow/control flow/modular design

(For Academic Oriented Projects)

3.1. System Design (*revised from ADD*)

3.1.1. System Model

3.1.2. Flowchart and/or pseudo code for proposed algorithms

3.1.3. Comparison metrics

3.1.4. Data set or benchmarks

3.2. System Architecture (*revised from ADD*)

Main aspects of data flow/control flow

4. Technical Approach and Implementation Details

This chapter includes *a detailed extension* of the Methodology and Technical Approach Section of the PSD document where you should put all details of your technical approach and implementation. You should emphasize hardware/software requirements of your project (as in the PSD) and you should write down all hardware components (such as Arduino, a special purpose kit etc.) and tools (including parsers, packages, programming languages such as Python, Scikit, Matlab and simulation tools such as Modelsim, Logisim, OMNeT++, Simul8, etc.) used in your project. Put a small picture for each hardware component used and explain each tool and hardware briefly by presenting why you selected them in your project.

Since this course is your main design project, you should also write down *all details* of the followings as part of this section.

- Data Structures used in your project
- Algorithms implemented in your project
- Operating System related details (such as multithreading, mutual exclusion, etc)
- Networking related details (such as socket programming, protocol definitions, etc)
- Finite State Machines (if used)
- Any other design/implementation detail from your project that was related with the courses you have attended during your undergraduate study.

Note that you should put all necessary subchapters within this chapter.

5. Software Testing / Experimental Study

Note that the title of this section will be different based on the type of the project. If the project is software oriented, title of this chapter will be “*Software Testing*”. If it is academic oriented then the title will be “*Experimental Study*”

As part of this chapter, you can add as many subchapters as you want. In this chapter, implementation and test phases of the project are described in detail. Definition of test tools, the details of experimental study and evaluation of performance metrics should be given in this part. Applied tests and scenarios should be defined and described clearly. Results of tests/scenarios are given as part of the section. It may also include a comparison study with

related work. Program codes, sample input data, sample output data can be attached to the report as an appendix.

6. Benefits and Impact (*revised from PSD*)

Benefits/Implications: What are the potential benefits of your project? Who will benefit from your project after its successful completion, and how?

Additionally, you should answer all four types of impacts listed below. In case of any of them that does not apply to your project, indicate with an explanation.

i. Scientific Impact: What would be the scientific impact of your project. Do you expect that it will be published in a scientific paper?

ii. Economic/Commercial/Social Impact: What type of outcome(s) is obtained and expected to be obtained (in near future) from the project: A (commercial) product? A prototype? A useful model? Startup company? Potential of import substitution? Media influence? Increase in life quality? Improve in education level? Contribution to sustainable environment and energy?

iii. Potential Impact on New Projects: Do you expect that this project will have a pioneering effect for future projects? Please describe briefly.

iv. Impact on National Security: Cyber security, energy security, border security, food security, etc. (if exists)

7. Conclusion and Future Work

This chapter will summarize the project briefly and present the results of the experiments or presents the results of the testing for the generated software architecture. The advantages and disadvantages of the solution/methods applied in the project are briefly described. Any future work considerations will be given as part of this chapter.

GENERAL FORMAT of the THESIS

Margins

For all pages of the thesis, the left hand margin will be 3.5 cm (because of binding) and all other page margins will be 2.5 cm.

Line Spacing, Justification and Indents

All text in abstract, acknowledgements, dedication and body of the text is 1,5 spaced. Table of Contents and Lists of Figures and Tables are single-spaced. All text in paragraph form is fully justified. First line indents are 0.75 cm.

Font

All text of the thesis is Times New Roman, 12 pt. For tables, figure or chart explanations font type may be other than Times New Roman (*e.g.* Arial) and size must be at least 8 point.

Equations

All equations should be indented and numbered by chapter and equation number consecutively; for example, “(Equation 4.1)”, where the number 4 represents the chapter number and 1 indicates that this is the first equation within chapter 4. The parentheses are required and equation numbers should be right justified.

Headings

All chapter titles and headings should follow these examples: (note that each chapter starts a new page).

1. CHAPTER TITLE (Upper Case, Bold, 14 Point, Left-justified)

1.1 First Level Heading (Lower Case, Bold, 12 Point, Indented 0.6 cm)

1.1.1 Second Level Heading (Lower Case, Normal, Underlined, 12 Point, Indented 1.2 cm)

Page Numbering

All pages, except the title page, are numbered in the lower margin 1.25 cm from the bottom of the page and centered. Pagination begins with the Approval Page, which is page ii. All other pages up to body of the thesis are numbered in lower case Roman numbers. The body of the thesis will be numbered starting with 1 and continuing until the last page of the thesis.

Spelling

Spell check should be done before submission.

Figures and Tables

Figures/Tables should be numbered consecutively by chapter and figure/table number. For example, for Figure 4.2, the number 4 represents the chapter number and number 2 indicates it is the second figure within Chapter 4. The figure/table number should be followed by a brief, descriptive caption. If the figure/table is citation from any source, the source of the figure/table should be written to end of the title.

References

References in the text are given with the number of the corresponding reference in the reference section in brackets. Some examples from the text are [1], [7,12]. Note that the reference numbers in the text should appear in increasing order.

In the reference section, references should be based on the format given in the following examples.

Journal sample:

When listing journal articles in an enumerated list, the order of information should be as follows:

- The author's name (or names)(Surname, initials of the first author and the initial, surname for the rest),
- The title of the article (between quotation marks),
- The title of the journal (in italics),
- As much other information as you can find about the journal, for example the volume and issue numbers,
- The page number/s of the article in the journal(p for single page, pp for the range of pages),
- The year of publication.

In Reference list:

Ebrahimi, M., A. Evans, M. B. Tahoori, E. Costenaro, D. Alexandrescu, V. Chandra and R. Seyyedi, "Comprehensive Analysis of Sequential and Combinational Soft Errors in an Embedded Processor", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems , Vol. 34, No. 10, pp. 1586–1599, Oct 2015.

Conference Paper Sample:

When listing conference papers in an enumerated list, the order of information should be as follows:

- The author's name (or names)(Surname, initials of the first author and the initial, surname for the rest),
- The title of the conference paper (between quotation marks),
- In: followed by editor or name of organisation,
- Full title of conference (in italics),
- Location
- Date
- Publisher,
- Place of publication
- The year of publication

In Reference list:

Alameldeen, A. R., I. Wagner, Z. Chishti, W. Wu, C. Wilkerson and S.-L. Lu, "Energy-efficient Cache Design Using Variable-strength Error-correcting Codes", Proceedings of

the 38th Annual International Symposium on Computer Architecture , ISCA '11, ACM, New York, NY, USA, 2011.

Book sample:

When listing a book in an enumerated references list, the order should be as follows:

- The author's or editor's name (or names)(Surname, initials of the first author and the initial, surname for the rest),
- The title of the book (in italics),
- If it is an edition other than the first,
- The name of the publisher,
- The city the book was published in,
- The year the book was published

In Reference list:

Doebelin, E. O., Control System Principles and Design , John Wiley & Sons, Inc., New York, NY, USA, 1985.

Web page sample:

When listing websites in an enumerated list, the order of information should be as follows:

- The author's name (Surname, initials) or source,
- The title of web document or page (in italics),
- Year
- URL
- Accessed date

In Reference list:

Clark, C. *Physicists Crack Another Piece of The Glass Puzzle*. 2012,
<http://www.rdmag.com/news/2012/10/physicists-crack-another-piece-glass-p...>
Date accessed: 15 Aralık 2012.

Appendices

A cover page must precede each appendix with the appendix labelled by ascending letters (e.g. A, B, C, etc.) and title.